Dylan Costa

Website | dylan.costa@uconn.edu | LinkedIn

Research Interests

Algebraic Number Theory and Arithmetic Geometry, with a focus on elliptic curves, modular curves, and Galois representations attached to abelian varieties.

Education

University of Connecticut 2021 – 2027 *(expected)*
PhD in Mathematics | Advisor: Álvaro Lozano-Robledo

Ithaca College 2017 – 2021
BA in Mathematics, *with Honors* | Advisors: Emilie Wiesner & Ted Galanthay

Teaching Experience

Instructor of Record

MATH 2410: Elementary Differential Equations — Fall 2025, Spring 2026
2 sections of 40 students each semester

MATH 1071: Calculus for Business and Economics — Fall 2023, Spring 2024, Spring 2025
2 sections of 40 students each semester

Graduate Teaching Assistant

MATH 3160: Probability *(Online)* — Summer 2025, Summer 2026

MATH 2110: Multivariable Calculus — Summer 2024, Fall 2024
3 sections of 40 students each semester

MATH 1132: Calculus II — Spring 2022, Summer 2022, Fall 2022, Spring 2023, Summer 2023, Summer 2026
3 sections of 40 students each semester

MATH 1131: Calculus I — Fall 2026, Fall 2021
2 sections of 40 students each semester

Mentoring

Independent Study: Cryptography Spring 2026
University of Connecticut

Directed Reading Project in Algebraic Number Theory Fall 2025
University of Connecticut | Directed Reading Program

Directed Reading Project on Algebraic Curves Spring 2024
University of Connecticut | Directed Reading Program

Publications

ℓ-primary Torsion in Elliptic Curves over $\mathbb{Q}$ — In preparation
With Álvaro Lozano-Robledo and Enrique González-Jiménez.

Classification of Isogeny-Torsion Graphs in Quadratic Fields — In preparation
With Rakvi.

Rank Growth of Elliptic Curves in Quadratic Extensions — In preparation
With Álvaro Lozano-Robledo.

An Elliptic Curve of Rank ≥ 34 *and Root Number* -1 *over a Quadratic Field* — October, 2026
With Álvaro Lozano-Robledo. Preprint

Invited Talks

Recent Discoveries on Ranks of Elliptic Curves
SIGMA Graduate Seminar, University of Connecticut — October 2026

A Computational Perspective on Modular Curves
Number Theory Reading Group, University of Connecticut — November 2025

Rank Growth of Elliptic Curves over Quadratic Extensions
SIGMA Graduate Seminar, University of Connecticut — August 2025

Elliptic Curves and their Applications in Mathematics and Beyond
Mathematics Colloquium, Ithaca College — April 2025

Pontryagin Duality and Applications to Tate's Thesis
Number Theory Reading Group, University of Connecticut — March 2025

Rank Growth of Elliptic Curves over Quadratic Extensions
Graduate Research Forum, University of Connecticut — January 2025

On ℓ-adic Images of Galois Attached to Elliptic Curves over the Rationals
General Exam Presentation, University of Connecticut — November 2024

On the Euclidean Embedding and Adeles/Ideles of a Number Field
Number Theory Reading Group, University of Connecticut — October 2024

Construction of High Rank Elliptic Curves over Quadratic Fields
Number Theory Reading Group, University of Connecticut — April 2024

Elliptic Curves and their Applications
Mathematics Continued Conference, University of Connecticut — March 2024

On Meromorphic Differentials and Riemann-Roch
Number Theory Reading Group, University of Connecticut — April 2023

Awards

Summer Travel Fellowship — Summer 2026
University of Connecticut

Oracle Honors Society — Fall 2019 – Spring 2021
Ithaca College

Summer Scholar Fellowship — Summer 2020
Ithaca College

Ramanujan Award in Mathematics — Fall 2019
Ithaca College

Relevant Skills

Programming MAGMA, SAGE, Java *(advanced)*, Python, MATLAB *(intermediate)*, HTML *(novice)*
Software LaTeX, Excel, Google Sheets

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